Dataset Memo: Flickr30k Subset for Text-to-Image Generation

IE 7615 Deep Learning Architectures · Project 2, Milestone 1 · Group 02: Abdellah Faleh, Jin-Woo Hong, Samuel Tong, Tristan Lyons · October 20, 2026

Dataset choice and size

We use **2,000 photos with captions** chosen at random from the **Flickr30k** dataset. Flickr30k has 31,783 everyday photos, and each photo has 5 captions written by people. We download it from Hugging Face (lmms-lab-encoder/flickr30k) and only download the rows we need, not the full 4.4 GB.

Group	Pairs	Used for
Train	1,600	Fine-tuning (if we do it)
Validation	200	Choosing settings
Test	200	Final scores and the image gallery

Why Flickr30k? People wrote the captions while looking at each photo, so the text matches the image well. Other options, like Conceptual Captions and LAION, use text copied from websites, which often does not match the image. 2,000 pairs fit the assignment's range (1,000 to 5,000) and take about 10 minutes to prepare on Google Colab.

How we prepared the data

We keep a photo and its caption only if they pass every check below, and stop at 2,000 good pairs. All settings are saved in *dataset_config.json* with a fixed random seed (42), so anyone can rebuild the exact same dataset.

1. **Clean the captions:** fix extra spaces ("dog ." becomes "dog"), remove the final period, start with a capital letter.
2. **Check caption length:** keep 6 to 30 words. Short captions are too vague; CLIP reads at most 77 word pieces.
3. **Check caption quality:** remove captions with a question mark (the writer was unsure), captions that are mostly symbols, and captions with words about nudity, violence or weapons.
4. **Pick one caption per photo:** keep the good caption closest to 15 words; save the others as extras.
5. **Check the photo:** remove photos under 256 pixels on the short side, and very wide or tall photos (over 2:1).
6. **Remove repeats:** remove repeated captions, and photos that look almost the same as one we already have (compared with an image "fingerprint").
7. **Resize:** cut a square from the center and resize to 512×512 pixels, the size Stable Diffusion uses.
8. **Normalize:** change pixel values from 0 to 255 into the range -1 to 1, the format the image generator expects.
9. **Final check:** the notebook checks that there are 2,000 unique pairs, all 512×512 color images.

Cleaning results. We checked 2,031 random photos to get 2,000 good pairs (98.5% passed):

Step	Number of photos
Photos checked	2,031
Removed: photo too small or wrong shape	30
Removed: no good caption	1
Removed: repeated caption or photo	0
Kept	**2,000**

Of about 10,000 captions checked, 95% passed (removed: 378 too short, 65 too long, 34 unsafe words, 6 unsure, 1 mostly symbols). Kept captions are 14 words on average (7 to 23). Average color on a 0 to 1 scale: red 0.441, green 0.409, blue 0.371.

License

Flickr30k does not have an open license. The photos still belong to the people who posted them on Flickr. The dataset creators allow use **only for non-commercial research and education**, following Flickr's Terms of Use. Our class project is non-commercial education, so our use is allowed. Because of this:

- We **do not upload the 2,000 photos to GitHub** (only a few appear in our comparison figures). The repository has the notebook and caption file, so anyone can download the same photos again.

- We use the photos only for this course.
- We cite the dataset: Young, Lai, Hodosh and Hockenmaier, *From image descriptions to visual denotations*, TACL 2014.
- Stable Diffusion 1.5 has its own separate license (CreativeML OpenRAIL-M).

What the photos show

Most Flickr30k photos show **people doing everyday things**: walking, playing sports, working, eating. Animals, streets, water and nature also appear. There are very few close-up objects, no drawings or artwork, and almost no images with text, so our results may not apply to those kinds of images.

Topic	People	Animals	Sports	Water / beach	City / street	Nature	Indoor
% of captions	70.2	12.2	15.5	7.8	10.4	13.4	5.8

Because most photos show people, this is a hard test for Stable Diffusion. Image generators often make mistakes with faces, hands, body positions and the number of people ("three men"). Our 15 test captions include every topic so we can see these mistakes.

Ethics

- **Real people.** The photos show real people who did not agree to be used for AI research. We only use them to compare with generated images, never to recreate a specific person, and we do not share them.
- **Biased captions.** Some captions guess a person's race, age or gender from how they look (for example "an Asian man"). These guesses can be wrong and can teach a model stereotypes. We keep the original captions but point out these cases in our review.
- **Mostly Western photos.** Most Flickr users at that time were from North America and Europe, so places, clothes and activities come mostly from those regions.
- **Bias in the model.** Stable Diffusion learned from internet images and can repeat stereotypes, for example showing only men for some jobs. We keep its safety filter on, removed unsafe captions, and will report biased results honestly.
- **Possible misuse.** Realistic images of people can be misused. We use generated images only for this course and always label them as AI-generated.